



Cyber Security Internship Programme

Lab Setup Report

Building a Penetration Testing Lab: Kali Linux Virtual Machine

The goal of this task was to build a personal penetration-testing lab using a virtualization platform with Kali Linux as the guest machine. An existing Kali Linux VM (set up previously) was used for this task: it was booted up, the desktop environment was explored, and basic verification commands were run (**whoami**, **ip a**, **sudo apt update**) to confirm the system was fully operational and network-connected.

Intern Name Mahnoor Chaudhary	Task Week 1 · Task 3
System Username mahnoor@kali	Status Completed
Program Cyber Security Internship	Deadline Aug 24, 2026 · 10:00 PM

1. Lab Environment

Component	Detail
Virtualization Platform	VMware Workstation
Guest OS	Kali Linux 2026.06 (kernel 7.0.12-2kali1, x86_64)
Allocated Resources	4 GB RAM · 2 vCPUs · 100 GB Disk
Network Mode	NAT / Host-only adapter (eth1: 192.168.61.129/24)

2. Installation Steps Performed

1. Verified that the virtualization platform (VMware Workstation) and the previously set-up Kali Linux VM were correctly configured, with 4 GB RAM, 2 vCPUs, and 100 GB of disk space allocated.
2. Powered on the existing Kali VM and logged in.
3. Opened a terminal (mahnoor@kali) and explored the desktop environment.
4. Verified the system by running `whoami`, `uname -a`, and `ip a`.
5. Ran `sudo apt update` to confirm internet connectivity and refresh package repositories.
6. Completed the TryHackMe "Linux Fundamentals Part 1" room, practicing core navigation commands directly in the Kali terminal.

3. Screenshots — Environment & Verification

3.1 Kali Linux Desktop Environment

The existing Kali Linux VM was booted up and the desktop environment loaded with the default Kali wallpaper, taskbar, and application shortcuts, confirming the system is set up correctly and ready to use.



Figure 1 — Kali Linux desktop running inside the virtual machine.

3.2 System Identity Check

`uname -a` was used to confirm the kernel version and system architecture, verifying the VM is running the Kali 7.0.12 kernel on x86_64.

```
mahnoor@kali: ~/Desktop
Session Actions Edit View Help
mahnoor@kali)~$ uname -a
Linux kali 7.0.12-kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.0.12-2kali1 (2026-06-18) x86_64 GNU/Linux
mahnoor@kali)~$
```

Figure 2 — Terminal output of 'uname -a' confirming the Kali kernel and hostname.

3.3 Network Configuration

`ip a` confirmed two active network interfaces: the loopback interface (lo) and eth1, which was assigned the address 192.168.61.129/24, verifying the VM has working network connectivity.

```
mahnoor@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:58:3f:ec brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:58:3f:60 brd ff:ff:ff:ff:ff:ff
    inet 192.168.61.129/24 brd 192.168.61.255 scope global dynamic noprefixroute eth1
        valid_lft 1554sec preferred_lft 1554sec
    inet6 fe80::20c:29ff:fe58:3ff6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 3 — Terminal output of 'ip a' showing active network interfaces.

4. TryHackMe — Linux Fundamentals (Part 1)

The free-tier TryHackMe room "Linux Fundamentals Part 1" was completed as required proof of practical work. The room covered navigating the Linux filesystem and using core shell commands directly inside the Kali terminal. The commands practiced are summarised below.

Command	Purpose
<code>pwd</code>	Print the current working directory.
<code>ls</code>	List files and folders in the current directory.
<code>cd</code>	Change the current directory / navigate the filesystem.
<code>cat</code>	Display the contents of a file directly in the terminal.
<code>echo</code>	Print text or variable values to the terminal output.
<code>whoami</code>	Display the username of the currently logged-in user.

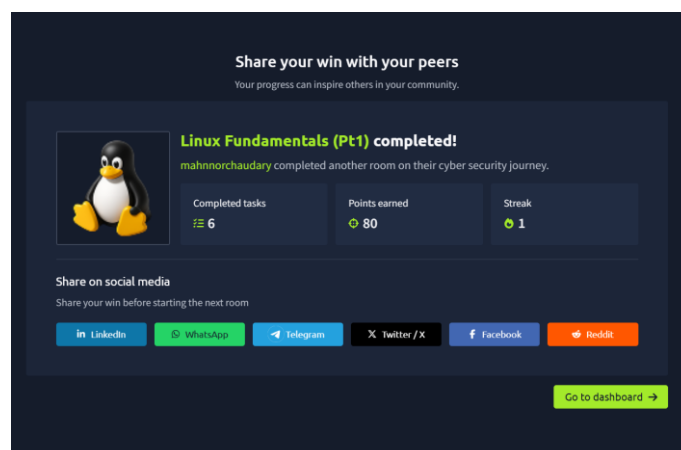


Figure 4 — Completion badge: 'Linux Fundamentals (Pt1)' — 6 tasks completed, 80 points earned.

Result: Room completed with 6/6 tasks finished and 80 points earned, confirming hands-on practice of the Linux commands listed above.

5. Errors / Issues Faced

No blocking errors were encountered while booting the existing Kali Linux VM. The system started up cleanly, network connectivity was available immediately on eth1, and all verification commands executed successfully on the first attempt.

6. Conclusion

The penetration-testing lab environment was confirmed ready with Kali Linux running as a virtual machine. System identity, kernel version, and network configuration were verified via the terminal, and the TryHackMe "Linux

Fundamentals Part 1" room reinforced core command-line skills (**pwd**, **ls**, **cd**, **cat**, **echo**, **whoami**). The lab is ready for use in upcoming reconnaissance and exploitation exercises.